SYSTEM ENGINEER | PUZZLES

Infosys
Puzzle Solving

50 Puzzle-Based Practice Questions

Weighing, crossing, Sudoku, cryptarithm, logic and scheduling puzzles.

50	4	100%
Questions	Options each	With solutions

WHAT'S INSIDE

Weighing Puzzles
4 questions
Balances, weights, coins

Crossing Puzzles
4 questions
Bridge, river and boat

Sudoku & Latin Grids
5 questions
Digit and letter grids

Cryptarithmetic
5 questions
Letters standing for digits

Magic Squares & Grids
4 questions
Sums in rows and columns

Truth & Lies
5 questions
Knights, knaves, liars

Scheduling & Order
7 questions
Days, rows, races, stacks

Measuring & Time
4 questions
Jugs, ropes, switches, clocks

Mathematical Puzzles
12 questions
Counting, divisors, pigeonhole

HOW TO USE

Give each question about 2 minutes, then check the answer and the solution below it.

Infosys Puzzle Solving

50 Puzzle Questions

1 You have 10 bags of coins. Every coin in 9 bags weighs 10 g, but every coin in one bag weighs 11 g. What is the minimum number of weighings on a digital scale needed to find the heavier bag?

- A) 2
- B) 1
- C) 3
- D) 4

Answer: B) 1

Take 1 coin from bag 1, 2 from bag 2, and so on up to 10 from bag 10, and weigh them together. The excess weight in grams equals the number of the heavier bag. So one weighing is enough.

2 Four people take 1, 2, 5 and 10 minutes to cross a bridge. Only two can cross at a time with one torch, at the slower person's speed. What is the minimum total time for all to cross?

- A) 17
- B) 15
- C) 19
- D) 21

Answer: A) 17

Send 1 and 2 across (2 min), 1 returns (1 min), 5 and 10 cross (10 min), 2 returns (2 min), then 1 and 2 cross (2 min). Total = 2 + 1 + 10 + 2 + 2 = 17 minutes.

3 Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box has the digits 1 to 4. What is the sum of the missing digits?

2	?	?	3
?	3	2	?
4	?	3	?
?	2	?	1

- A) 18
- B) 20
- C) 19
- D) 21

Answer: B) 20

The completed rows are 2 4 1 3, 1 3 2 4, 4 1 3 2 and 3 2 4 1. The missing digits are 4, 1, 1, 4, 1, 2, 3 and 4, which add up to 20.

Infosys Puzzle Solving

50 Puzzle Questions

4 In the cryptarithm BASE + BALL = GAMES, each letter stands for a different digit. What is the value of G + A + M + E + S?

- A) 23
- B) 25
- C) 24
- D) 26

Answer: B) 25

7483 + 7455 = 14938. So G = 1, A = 4, M = 9, E = 3 and S = 8, and their sum is 25.

5 This 3 × 3 grid is a magic square. Find the number in the middle-right cell.

12	?	10
7	9	?
?	13	6

- A) 9
- B) 10
- C) 12
- D) 11

Answer: D) 11

The completed square is 12 5 10 / 7 9 11 / 8 13 6 with a magic sum of 27. Middle row: 27 - 7 - 9 = 11.

6 On an island, knights always tell the truth and knaves always lie. A says, 'We are both knaves.' What are A and B?

- A) A is a knight, B is a knave
- B) Both are knights
- C) A is a knave, B is a knight
- D) Both are knaves

Answer: C) A is a knave, B is a knight

If A were a knight, the statement would be true, which is impossible. So A is a knave and the statement is false, which means B is not a knave. B is a knight.

Infosys Puzzle Solving

50 Puzzle Questions

7 Five meetings P, Q, R, S and T are on different days from Monday to Friday. P is the day after Q. R is not on Monday or Friday. S is on Friday and T is two days before S. Which meeting is on Thursday?

- **A** P
- **B** Q
- **C** T
- **D** R

Answer: D) R

S is on Friday and T on Wednesday. P must follow Q, which fits only Monday and Tuesday, so Q is on Monday and P on Tuesday. That leaves R on Thursday.

8 You have a 7-litre jug and a 4-litre jug and unlimited water. Each fill, empty or pour counts as one step. What is the minimum number of steps to measure exactly 5 litres?

- **A** 8
- **B** 6
- **C** 7
- **D** 9

Answer: A) 8

Fill 4 and pour into 7 (twice, leaving 1 L in the 4-litre jug), empty the 7, move the 1 L into it, fill 4 and pour it in. The 7-litre jug holds 1 + 4 = 5 L after 8 steps.

9 100 lockers are closed. Person 1 toggles every locker, person 2 every 2nd locker, person 3 every 3rd, and so on up to person 100. How many lockers are open at the end?

- **A** 10
- **B** 25
- **C** 50
- **D** 100

Answer: A) 10

A locker is toggled once for each of its divisors. Only perfect squares have an odd number of divisors, so only they stay open. There are 10 perfect squares up to 100.

Infosys Puzzle Solving

50 Puzzle Questions

10 Using weights of 1 kg, 3 kg, 9 kg and 27 kg on both pans of a balance, what is the maximum weight up to which every whole-kg weight can be measured?

- A) 40 kg
- B) 27 kg
- C) 36 kg
- D) 41 kg

Answer: A) 40 kg

Weights in powers of 3 can be placed on either pan or left out, which measures every whole number up to their total. 1 + 3 + 9 + 27 = 40 kg.

11 A farmer takes a wolf, a goat and a cabbage across a river in a boat that carries only him and one item. The wolf cannot be left with the goat, nor the goat with the cabbage. What is the minimum number of one-way crossings?

- A) 5
- B) 9
- C) 7
- D) 11

Answer: C) 7

Take the goat over, return alone, take the wolf over, bring the goat back, take the cabbage over, return alone, then take the goat over. That is 7 one-way crossings.

12 Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box has the digits 1 to 4. Which digit replaces the ? in the second row?

3	?	4	?
?	2	?	1
1	?	2	?
?	4	?	3

- A) 1
- B) 4
- C) 2
- D) 3

Answer: B) 4

The completed rows are 3 1 4 2, 4 2 3 1, 1 3 2 4 and 2 4 1 3. The ? in the second row is 4.

Infosys Puzzle Solving

50 Puzzle Questions

13 In FORTY + TEN + TEN = SIXTY, each letter stands for a different digit. What is the value of S + I + X + T + Y?

- A) 20
- B) 22
- C) 21
- D) 24

Answer: B) 22

29786 + 850 + 850 = 31486. So S = 3, I = 1, X = 4, T = 8 and Y = 6, and their sum is 22.

14 In a 2 × 2 grid, the row sums are 7 and 11 and the column sums are 8 and 10. If the top-left number is 3, what is the bottom-right number?

- A) 4
- B) 6
- C) 5
- D) 7

Answer: B) 6

The top-left is 3, so the top-right = 7 - 3 = 4 and the bottom-left = 8 - 3 = 5. The bottom-right = 11 - 5 = 6.

15 P says, 'Q is a liar.' Q says, 'R is a liar.' R says, 'P and Q are both liars.' Each person is either always truthful or always lying. Who is truthful?

- A) P
- B) R
- C) Nobody
- D) Q

Answer: D) Q

If Q is truthful, R is a liar, so R's statement is false. P says Q is a liar, which is false, so P is a liar. This is consistent, so only Q is truthful.

16 Ann, Ben and Cara own a cat, a dog and a parrot, one each. Ann does not own the dog, Ben owns neither the cat nor the dog, and Cara does not own the parrot. Who owns the dog?

- A) Ann
- B) Cara
- C) Ben
- D) Cannot be determined

Answer: B) Cara

Ben owns neither the cat nor the dog, so Ben owns the parrot. Ann does not own the dog, so Ann owns the cat and Cara owns the dog.

17 Two ropes each burn for exactly 60 minutes, but unevenly. Rope 1 is lit at both ends and rope 2 at one end at the same time. When rope 1 burns out, rope 2's other end is lit. When does rope 2 burn out?

- A) 45 minutes
- B) 30 minutes
- C) 50 minutes
- D) 60 minutes

Answer: A) 45 minutes

Rope 1 burns out in 30 minutes. Rope 2 then has 30 minutes of burning left, and lighting its other end halves that to 15 minutes. Total = 30 + 15 = 45 minutes.

18 A snail climbs 5 m up a 20 m well during the day and slips 3 m back at night. On which day does it reach the top?

- A) 8th day
- B) 10th day
- C) 11th day
- D) 9th day

Answer: D) 9th day

Each full day and night gains 2 m, so after 8 days the snail is at 16 m. On the 9th day it climbs 5 m and reaches 21 m, which is past the top.

19 Three boxes A, B and C are weighed in pairs: A + B = 70 kg, B + C = 80 kg and A + C = 90 kg. What is the weight of the heaviest box?

- A) 40 kg
- B) 45 kg
- C) 50 kg
- D) 60 kg

Answer: C) 50 kg

Adding all three gives 2(A + B + C) = 240, so A + B + C = 120. C = 120 - 70 = 50 kg, A = 40 kg and B = 30 kg. The heaviest box is 50 kg.

20 Four people take 2, 3, 6 and 10 minutes to cross a bridge with one torch. At most two can cross together, at the slower person's speed. What is the minimum total time?

- A) 21
- B) 19
- C) 20
- D) 23

Answer: A) 21

Send 2 and 3 across (3 min), 2 returns (2 min), 6 and 10 cross (10 min), 3 returns (3 min), then 2 and 3 cross (3 min). Total = 3 + 2 + 10 + 3 + 3 = 21 minutes.

21 Fill the 4 × 4 Sudoku so that every row, column and 2 × 2 box has the digits 1 to 4. Which digit replaces the ? in the last row?

1	?	?	3
?	4	1	?
2	?	3	?
?	3	?	1

- A) 1
- B) 3
- C) 4
- D) 2

Answer: D) 2

The completed rows are 1 2 4 3, 3 4 1 2, 2 1 3 4 and 4 3 2 1. The ? in the last row is 2.

22 In EAT + THAT = APPLE, each letter stands for a different digit. What is the value of APPLE?

- A) 10083
- B) 10038
- C) 10308
- D) 10380

Answer: B) 10038

819 + 9219 = 10038. So A = 1, P = 0, L = 3 and E = 8, which makes APPLE = 10038.

23 Fill 1 to 9 once each so that every row, column and diagonal adds to 15. Find the number at the ?

4	9	2
3	5	?
?	?	?

- A) 8
- B) 6
- C) 7
- D) 9

Answer: A) 8

The only valid square is 4 9 2 / 3 5 7 / 8 1 6. So the bottom-left cell is 8.

24 Two guards stand at two doors, and one door leads to treasure. One guard always lies and the other always tells the truth. Which single question to either guard lets you find the treasure door?

- A) Ask: are you the truthful one?
- B) Ask: is the left door correct?
- C) Ask: what would the other say?
- D) Ask: which of you is the liar?

Answer: C) Ask: what would the other say?

Whichever guard you ask, the answer points to the wrong door: the liar misreports the truthful guard's answer, and the truthful guard reports the liar's false answer. So you pick the other door.

Infosys Puzzle Solving

50 Puzzle Questions

25 Six people A, B, C, D, E and F sit in a row facing north. C is second from the left, F is at the right end, and A is immediately right of C. E is not next to F, and B is immediately left of D. Who sits at the left end?

- A) A
- B) B
- C) E
- D) D

Answer: C) E

The order from the left is E, C, A, B, D, F. So E sits at the left end.

26 Three switches outside a closed room control three bulbs inside. You may enter the room only once. What is the minimum number of entries needed to match every switch to its bulb?

- A) 2
- B) 3
- C) 1
- D) 4

Answer: C) 1

Turn switch 1 on for a few minutes, then off, then turn switch 2 on and enter. The lit bulb is switch 2, the warm unlit bulb is switch 1 and the cold bulb is switch 3. One entry is enough.

27 Two trains 100 km apart move towards each other at 30 km/h and 20 km/h. A bird flies back and forth between them at 60 km/h until they meet. How far does the bird fly?

- A) 100 km
- B) 150 km
- C) 180 km
- D) 120 km

Answer: D) 120 km

The trains meet after 100 ÷ (30 + 20) = 2 hours. The bird flies for 2 hours at 60 km/h, so it covers 120 km.

28 On a balance, 3 apples weigh the same as 2 mangoes, and 4 mangoes weigh the same as 6 oranges. How many oranges weigh the same as 6 apples?

- A) 4
- B) 6
- C) 8
- D) 9

Answer: B) 6

Doubling the first statement, 6 apples = 4 mangoes. Since 4 mangoes = 6 oranges, 6 apples = 6 oranges.

29 Three missionaries and three cannibals cross a river in a boat holding at most two people. Cannibals may never outnumber missionaries on a bank. What is the minimum number of one-way crossings?

- A) 9
- B) 11
- C) 13
- D) 15

Answer: B) 11

This is a classic puzzle. Checking every safe arrangement on both banks shows that the shortest safe plan needs 11 one-way crossings.

30 Fill the 3 × 3 grid with the letters A, B and C so that each row and column has every letter once. Which letter replaces the ?

A	?	?
?	C	?
?	?	?

- A) A
- B) B
- C) C
- D) Cannot be determined

Answer: B) B

The only valid fill is A B C / B C A / C A B. So the bottom-right cell is B.

Infosys Puzzle Solving

50 Puzzle Questions

31 In HERE + SHE = COMES, each letter stands for a different digit. What is the value of COMES?

- A) 10438
- B) 10384
- C) 10843
- D) 10348

Answer: D) 10348

9454 + 894 = 10348. So C = 1, O = 0, M = 3, E = 4 and S = 8, which makes COMES = 10348.

32 In a calendar month, a 2 × 2 block (two dates in a row and the two dates directly below them) adds up to 64. What is the smallest date in the block?

- A) 10
- B) 14
- C) 12
- D) 16

Answer: C) 12

If the smallest date is d, the block is d, d + 1, d + 7 and d + 8, which add up to 4d + 16. So 4d + 16 = 64 and d = 12.

33 A says, 'B is a knight.' B says, 'A and I are of opposite types.' Knights always tell the truth and knaves always lie. What are A and B?

- A) Both are knaves
- B) A is a knight, B is a knave
- C) A is a knave, B is a knight
- D) Both are knights

Answer: A) Both are knaves

If A were a knight, B would be a knight, but then B's statement would be false, which is impossible. So A is a knave, which makes B a knave too, and B's false statement fits.

34 Five runners A, B, C, D and E finish a race with no ties. E finishes first and C finishes last. B finishes before A, and D finishes immediately after B. Who finishes third?

A) D

B) A

C) B

D) C

Answer: A) D

E is 1st and C is 5th. B, D and A fill 2nd to 4th, with B before A and D right after B. The order is E, B, D, A, C, so D finishes third.

35 A clock is set correctly at 12 noon and gains 5 minutes every hour. What time will it show when the correct time is 6 PM?

A) 6:30 PM

B) 6:25 PM

C) 6:35 PM

D) 7:00 PM

Answer: A) 6:30 PM

In 6 hours the clock gains 6 × 5 = 30 minutes. So it shows 6:30 PM when the correct time is 6 PM.

36 What is the maximum number of pieces a round cake can be cut into with 4 straight cuts?

A) 8

B) 11

C) 10

D) 16

Answer: B) 11

The maximum number of pieces with n cuts is $n(n + 1)/2 + 1$. For $n = 4$: $4 \times 5/2 + 1 = 11$.

Infosys Puzzle Solving

50 Puzzle Questions

37 Place the digits 1, 2, 3 and 4 once each in a 2 × 2 grid so that the top row adds to 3, the bottom row adds to 7 and the left column adds to 4. What is the bottom-right digit?

- A 1
- B 2
- C 3
- D 4

Answer: D) 4

The top row must be 1 and 2, and the bottom row 3 and 4. The left column adds to 4, so it is 1 and 3. That makes the top-right 2 and the bottom-right 4.

38 In GO + TO = OUT, each letter stands for a different digit. What is the value of OUT?

- A 102
- B 120
- C 201
- D 210

Answer: A) 102

81 + 21 = 102. So G = 8, O = 1, T = 2 and U = 0, which makes OUT = 102.

39 Three people speak. A: 'Exactly one of us is lying.' B: 'Exactly two of us are lying.' C: 'All three of us are lying.' Who is telling the truth?

- A A
- B C
- C Nobody
- D B

Answer: D) B

C cannot be truthful, since that would make C a liar. If A were truthful, B and C would both be lying, which is two liars, not one. If B is truthful, A and C are the two liars, which fits.

Infosys Puzzle Solving

50 Puzzle Questions

40 Four boxes red, blue, green and yellow are stacked one above another. Yellow is at the bottom, green is immediately above yellow, and red is above blue. Which box is at the top?

- A Blue
- B Green
- C Yellow
- D Red

Answer: D) Red

From the bottom, yellow is first and green is second. Red is above blue, so blue is third and red is on top.

41 A knockout tournament has 32 teams, and every match eliminates one team. How many matches are played in total?

- A 16
- B 32
- C 31
- D 63

Answer: C) 31

To leave one champion, 31 teams must be eliminated, and each match eliminates exactly one team. So 31 matches are played.

42 Among four friends, A is taller than B, C is shorter than B, and D is taller than A. Who is the shortest?

- A A
- B B
- C D
- D C

Answer: D) C

The order from the tallest is D, A, B, C. So C is the shortest.

43 At a gathering, every two guests shake hands exactly once, and there are 45 handshakes in total. How many guests are there?

- A) 10
- B) 9
- C) 11
- D) 12

Answer: A) 10

With n guests, the handshakes = n(n - 1)/2. Setting this equal to 45 gives n(n - 1) = 90, so n = 10.

44 Three boxes are labelled 'Apples', 'Oranges' and 'Apples and Oranges', and every label is wrong. A fruit picked from the box labelled 'Apples and Oranges' is an apple. What does the box labelled 'Oranges' contain?

- A) Apples and oranges
- B) Only apples
- C) Only oranges
- D) Cannot be determined

Answer: A) Apples and oranges

The box labelled 'Apples and Oranges' is mislabelled, so it holds only apples. The box labelled 'Oranges' cannot hold oranges, and apples are taken, so it holds the mixed fruit.

45 What is the smallest positive integer that has exactly 6 divisors?

- A) 10
- B) 16
- C) 18
- D) 12

Answer: D) 12

12 has the divisors 1, 2, 3, 4, 6 and 12, which is exactly 6 divisors. No smaller number has exactly 6.

Infosys Puzzle Solving

50 Puzzle Questions

46 A two-digit number with non-zero digits is 4 more than 6 times the sum of its digits. What is the number?

- A) 46
- B) 54
- C) 64
- D) 74

Answer: C) 64

Let the number be 10a + b. Then 10a + b = 6(a + b) + 4, which gives 4a - 5b = 4. The only solution with non-zero digits is a = 6 and b = 4, so the number is 64.

47 A bag has marbles of 5 different colours, with plenty of each. What is the minimum number of marbles you must draw without looking to be sure of two of the same colour?

- A) 5
- B) 7
- C) 10
- D) 6

Answer: D) 6

In the worst case the first 5 marbles are all different colours. The 6th marble must match one of them, so 6 draws are needed.

48 A drawer has 10 white socks and 10 black socks. In the dark, how many socks must you pull out to be sure of having a pair of black socks?

- A) 3
- B) 11
- C) 12
- D) 20

Answer: C) 12

In the worst case you pull out all 10 white socks first. The next 2 socks must be black, so you need 10 + 2 = 12 socks.

Infosys Puzzle Solving

50 Puzzle Questions

49 What is the sum of all the digits used when writing the numbers from 1 to 50?

- **A** 315
- **B** 325
- **C** 330
- **D** 345

Answer: C) 330

The digits of 1 to 9 add up to 45. The numbers 10 to 49 add up to 55 + 65 + 75 + 85 = 280 in digits, and 50 adds 5. Total = 45 + 280 + 5 = 330.

50 How many squares of all sizes are there on a standard 8 × 8 chessboard?

- **A** 64
- **B** 140
- **C** 204
- **D** 264

Answer: C) 204

The number of k × k squares is $(9 - k)^2$. Total = $8^2 + 7^2 + 6^2 + 5^2 + 4^2 + 3^2 + 2^2 + 1^2 = 204$.

QUICK ANSWER KEY

Q1 to Q50

Q1 B	Q2 A	Q3 B	Q4 B	Q5 D	Q6 C	Q7 D	Q8 A	Q9 A	Q10 A
Q11 C	Q12 B	Q13 B	Q14 B	Q15 D	Q16 B	Q17 A	Q18 D	Q19 C	Q20 A
Q21 D	Q22 B	Q23 A	Q24 C	Q25 C	Q26 C	Q27 D	Q28 B	Q29 B	Q30 B
Q31 D	Q32 C	Q33 A	Q34 A	Q35 A	Q36 B	Q37 D	Q38 A	Q39 D	Q40 D
Q41 C	Q42 D	Q43 A	Q44 A	Q45 D	Q46 C	Q47 D	Q48 C	Q49 C	Q50 C